TECHNICAL ASSESSMENT

H2C Vacuum Debris Collection System

Pressure Sensors, Pneumatic Conveying, and Master's Thesis Research

Confirmed system correction

A separate pressure sensor is **already connected to the compressed-air pressure line**, although it is not shown in the supplied diagram. This report identifies that existing instrument as S_P, reporting P_{supply}. Retain it in its supply-monitoring role. There is **no need to add a supply sensor or repurpose S_1 or S_2 solely because S_P is absent from the photographs**.
The remaining selection questions concern S_2 for pickup vacuum and S_1 for discharge backpressure. The range, output, accuracy, and exact tapping location of the unshown S_P have not been provided; the photographed transmitters' labels must not be assigned to it.

Recommended direction. For this compact pickup-to-bag arrangement, establish a well-configured single-Line-Vac baseline, retain the existing S_P, use a documented vacuum-capable instrument at S_2, and use an appropriately sensitive low-pressure instrument at S_1. Retain the second Line Vac if testing demonstrates a worthwhile improvement in debris collection.

Basis and scope. This assessment uses the four-page supplied file `H2C_vacuum(1).pdf` [17], the user's clarification about S_P, manufacturer literature, and verified journal and engineering-method references. It is an engineering assessment of the proposed apparatus; actual pickup performance has not been measured. Product information, indicative U.S. prices, and research citations were reviewed on 17 October 2026.

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1 Updated interpretation of the apparatus

The photographs indicate the following debris-conveying sequence:

$$\text{Pickup} \longrightarrow S_2 \longrightarrow V_2 \longrightarrow V_1 \longrightarrow S_1 \longrightarrow \text{collection bag.}$$

Here, V_2 is the Line Vac nearest the pickup, and V_1 is the downstream Line Vac nearer the exhaust. Both units point in the same conveying direction. Their motive-air ports are supplied from the compressed-air line. The additional existing sensor S_P measures pressure somewhere on that supply line.

Table 1. The three existing pressure-measurement functions.

Instrument	Measurement role	Expected normal pressure	Revised treatment
S_P (existing; unshown)	Compressed-air supply pressure at its actual tapping point	Positive; the user reports an available supply of about 100 PSI	Retain and log during operation. Verify its own specifications and interpret its location.
S_2 (shown)	Suction pressure near the pickup, upstream of V_2	Below atmospheric pressure during suction	Use a documented vacuum or suitable compound-pressure sensor.
S_1 (shown)	Discharge pressure downstream of V_1, ahead of the collector	Usually near atmospheric pressure or modestly positive	Use a lower-range instrument for useful backpressure measurement.

The presence of S_P resolves the supply-sensor inventory question. It does not by itself establish adequate negative-pressure measurement at S_2 or adequate low-pressure sensitivity at S_1.

1.1 Pressure reference and location matter

Unless otherwise stated, pressure values in this report are gauge pressures relative to the local atmosphere:

$$p_g = p_{\text{absolute}} - p_{\text{atmosphere}}.$$

A negative gauge reading indicates suction. A positive supply pressure of 100 PSIG does not imply a vacuum of 100 PSI at the pickup.

The location of S_P remains relevant. If it is near the Line Vac inlet and downstream of restrictive components, it can closely represent the delivered motive pressure. If it is upstream of the regulator, coupling, or branch restrictions, it measures the upstream condition. This qualification does not imply that another permanent sensor must be purchased; first establish where the existing pressure tap is located.

2 Suitability of the photographed pressure sensors

2.1 The visible range does not establish vacuum capability

The label on the photographed transmitter appears to read:

- pressure range: **0–150 PSI**;
- electrical output: **0.5–4.5 V**;
- supply: **5 VDC**.

The manufacturer and exact part number are not legible enough to verify a datasheet. Assuming this is a conventional positive gauge-pressure transmitter, it is unsuitable for calibrated vacuum measurement at S_2. If the actual datasheet specifies a different pressure reference or a calibrated negative-pressure range, that information would change the assessment.

A transmitter may survive a vacuum without accurately measuring it. A negative value calculated in software can also arise from zero offset, noise, or extrapolation below the specified range. Therefore, use a sensor explicitly specified for vacuum, compound gauge pressure, or a suitable differential-pressure range.

Specifications of the unshown sensor are separate

The 0–150 PSI, 0.5–4.5 V, and 5 VDC markings above refer to the photographed transmitter type. The available information does not establish that the existing supply-line sensor S_P is the same model. Retain S_P and verify its own datasheet and calibration.

2.2 Why the range at S_1 also needs attention

S_1 is downstream of the ejectors. With a collection bag attached, its most useful role is tracking discharge resistance. A 0–150 PSI range is much wider than the small positive pressures of interest in this channel.

If S_1 remains upstream of bends and hose, its reading includes the effect of those downstream components as well as the bag. It is not automatically the pressure drop across the bag alone. To measure a filter's pressure drop specifically, measure on the two sides of that filter.

2.3 Proposed measuring ranges

Select the calibrated measuring range separately from the permissible overload or proof-pressure rating. A low-range sensor may have a high survival rating, but that does not extend its calibrated range.

Table 2. Initial range-selection approach; these are instrument ranges, not predicted operating pressures.

Channel	Range to consider	Selection rationale
Existing S_P	Verify its existing range; 0–150 or 0–200 PSIG is a typical candidate range for a 100 PSIG operating supply	Retain the installed instrument if its calibrated range and pressure ratings suit the actual supply. This is a verification criterion, not a new purchase recommendation.
S_2 pickup	Approximately −100 to 0 kPa gauge, or −100 to +100 kPa compound	A broad initial range while actual suction is being characterized. Narrow it later if small changes require better sensitivity.
S_1 discharge	Approximately 0 to 25 kPa gauge initially	More useful sensitivity for outlet resistance than a 0–150 PSI instrument.
Bag/filter differential	A narrower differential range selected from measurements	Resolve small pressure changes associated with filter loading.

3 One Line Vac versus two closely coupled units

3.1 What the manufacturer guidance supports

EXAIR states that two Line Vacs can improve conveying performance, but they do not double performance. Its guidance identifies the greatest benefit in longer conveying runs: a second unit can be located farther downstream to maintain particle velocity, including near a sweeping bend. It recommends spacing multiple units along the conveying path instead of grouping them together [1].

The two units in the supplied apparatus are close together. This arrangement may help, but the photographs cannot determine the additional pickup airflow, debris collection rate, or air consumption benefit. It should be compared with a correctly assembled single-unit arrangement.

Engineering recommendation: for a short pickup-to-bag application, first establish a good single-Line-Vac baseline. Keep the second unit if measured collection improvement justifies the added compressed-air demand and collector airflow.

3.2 Suction pressure is not a complete performance measurement

Debris collection depends on air entering the nozzle fast enough to entrain particles and maintaining sufficient transport through the hose.

An open pickup can move substantial air while showing modest negative pressure. Conversely, blocking the pickup can produce a more negative reading even though no pickup air or debris enters. A pressure reading alone cannot distinguish all useful and restricted-flow conditions.

For a selected section,

$$v = \frac{Q}{A}, \tag{1}$$

where v is mean air velocity, Q is actual volumetric airflow through that section, and A is its open area. Changing nozzle area also changes resistance and therefore Q. A smaller opening does not automatically improve collection and may increase plugging.

Use **collected debris mass per unit time** as the primary performance outcome, supported by pressure

and airflow measurements.

3.3 Illustrative vacuum and air-consumption data

Table 3 gives manufacturer catalog examples for standard aluminum Line Vacs. The exact model fitted to this apparatus could not be identified. All values are at **80 PSIG**, not 100 PSIG [3].

Table 3. Examples from the EXAIR catalog. kPa values are calculated from the listed inches of water.

Model	Nominal hose size	Motive-air consumption	Catalog vacuum
6081	1 inch	14.7 SCFM	-42 inH_2O; approximately -10.5 kPa
6083	$1\frac{1}{2}$ inches	33 SCFM	-36.8 inH_2O; approximately -9.2 kPa
6084	2 inches	45 SCFM	-28.5 inH_2O; approximately -7.1 kPa

These are catalog vacuum ratings, not guaranteed nozzle readings during debris pickup. SCFM in the table is **compressed-air consumption**; it is not pickup airflow. Do not scale the vacuum figures by 100/80 to predict operation at 100 PSIG.

For illustration only, two model 6083 units would have a combined catalog motive-air consumption of

$$Q_{\text{supply,total}} = 2(33) = 66 \text{ SCFM} \quad \text{at 80 PSIG.} \tag{2}$$

This is not a sizing result for the unidentified units in the photographs.

3.4 Use the existing S_P to assess the 100 PSI supply

EXAIR's aluminum Line Vac instructions cover normal shop-air supplies up to 100 PSIG and recommend checking pressure at the compressed-air inlet when flow or vacuum decreases [2].

Record the existing S_P readings with one unit operating and then with both operating. A no-flow pressure reading alone does not establish continuous air-delivery capability.

Interpret the result according to the actual tap location. If S_P is upstream of a restriction, a steady reading does not rule out pressure loss downstream. If needed, verify the pressure drop locally with suitable available instrumentation before deciding that more permanent sensors are required.

The photographs show a common supply manifold and a quick disconnect. Their presence warrants investigation, not a conclusion that they are undersized. EXAIR cautions that restrictive quick-connect and push-lock fittings can reduce performance [4]. The compressor, regulator, manifold, branches, and fittings must pass the required continuous air volume.

4 Collector, plumbing, and pressure-tap improvements

4.1 The collection bag must vent the conveying air

The bag is on the ejector discharge side. It receives the air drawn through the pickup plus the air injected by every active Line Vac. Conservation of air mass gives

$$\dot{m}_{\text{air,out}} = \dot{m}_{\text{air,pickup}} + \dot{m}_{\text{air},V_1} + \dot{m}_{\text{air},V_2}. \tag{3}$$

The term for the removed unit disappears in the single-unit arrangement. Volumetric flows can be added only after expressing them at the same reference pressure and temperature.

The engineering implication is that the second unit increases the airflow the bag must release, even if its pickup benefit is small. A restrictive collector can reduce the useful gain.

EXAIR describes Line Vacs as discharging toward ambient conditions and cautions against opposing discharge pressure and undersized exhaust piping [5]. Use a collector with sufficient breathable filtration area, an adequately sized inlet, and filtration suitable for the actual particle size.

A sealed plastic bag alone does not provide the required air outlet. It can serve as a liner within a suitably vented receiver, provided a filtered exhaust path remains. EXAIR documents a single-Line-Vac pellet-cleanup application using a large collection sack with a planned filtration path for the outgoing air [6]. This supports the architecture without establishing the appropriate model for the user's debris.

4.2 Changes to prioritize in the photographed assembly

1. **Simplify the outlet turn.** Test a straighter outlet or a larger-radius bend in place of the nearly 180-degree turn immediately after V_1. The existing geometry presents turning losses and possible debris-accumulation locations; the benefit of changing it should be measured.

2. **Preserve adequate discharge passage.** Avoid reducing the discharge to fit a small bag collar. Match the hose and connections to the manufacturer's specified conveyance size.

3. **Keep the active Line Vac near the pickup.** Use a short suction section and put most transport distance downstream. EXAIR favors mounting near the suction point and minimizing bends [2].

4. **Use a smooth nozzle transition.** Provide a passage appropriate for the largest expected debris and a controlled air-entry gap when collecting from a surface. Hold these dimensions fixed in the comparison test.

5. **Retain the existing supply-pressure instrumentation.** Keep S_P assigned to P_{supply}. Improve the local taps for S_1 and S_2 as needed without moving them onto the compressed-air supply line.

4.3 Updated schematics with the existing supply sensor

Figure 1 explicitly includes the existing unshown supply sensor. Figure 2 shows the recommended single-unit baseline. Both are functional schematics, not dimensioned plumbing drawings. The connection to S_P denotes its measurement role; its actual position relative to the regulator, coupling, and branches remains to be verified.

4.4 Pressure taps and debris protection

The existing S_1 and S_2 taps are located on bends. For more representative static-pressure measurements, use a straight suction section between the nozzle and the active Line Vac, and a straight discharge section toward the collector.

Use flush, burr-free sidewall openings perpendicular to flow rather than probes projecting into the stream. DwyerOmega explains that turbulence, aspiration, and impingement at a static tap can materially affect readings [7].

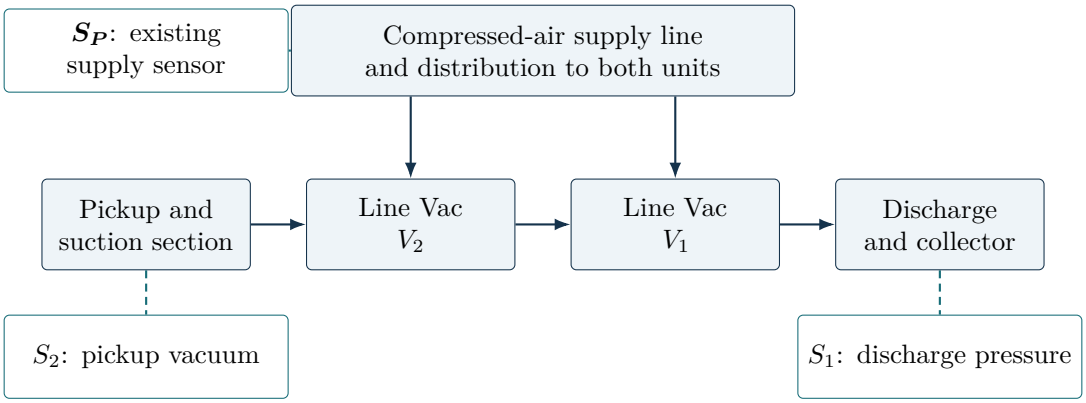


Figure 1. Existing tandem arrangement, including S_P confirmed by the user but absent from the supplied diagram. Dashed lines indicate pressure-measurement connections. The collector represents the intended downstream bag or receiver.

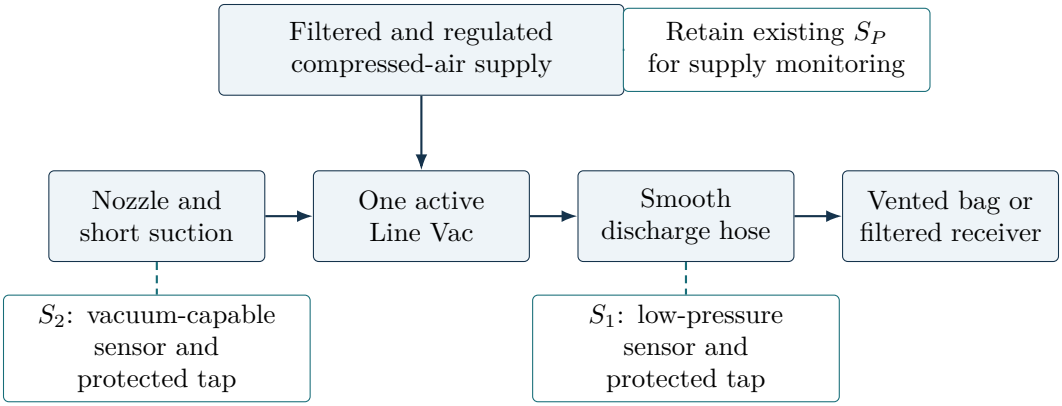


Figure 2. Recommended single-unit baseline. The existing S_P remains in its supply-monitoring role; the sensor-selection changes concern S_2 and S_1.

Place the transmitters outside the debris transport stream and use short sensing branches with accessible particulate protection suitable for vacuum service. A clogged branch can respond slowly or retain a stale pressure. Verify that the channel returns to atmospheric zero after shutdown.

A sensing-line dust filter is not an overpressure protector. Normal low pressure in the process duct also does not establish the allowable pressure of the printed parts or sensors under an abnormal blocked condition. Keep the collector vented and select pressure protection and component ratings for the actual arrangement.

5 Purchasable replacement sensors

The following candidates address S_2 and S_1. **No additional supply-pressure transmitter is included by default**, because S_P is already installed. Prices are indicative U.S. listings checked on 17 October 2026 and exclude cables, adapters, and interface electronics.

5.1 Candidates for S_2: pickup vacuum

Table 4. Vacuum-sensor candidates. Full purchasing links are included in the references.

Candidate	Measuring range	Output / supply	Price
ProSense `PTD25-20-VH`	−14.5 to 0 PSIG; approximately −100 to 0 kPa	4 mA at atmospheric gauge zero; 20 mA at full vacuum. 9.6–32 VDC supply.	$169
ProSense `EPS25-V14-1001`	−14.5 to +14.5 PSIG; approximately −100 to +100 kPa	Selectable 4–20 mA or 0–10 V, plus switching output. 18–30 VDC supply.	$360
Omega `PX309-015V5V`	0 to −15 PSIG vacuum	0–5 V output. 9–30 VDC supply.	$451.26

These ranges, outputs, and prices are supported by the product listings and manufacturer specifications [8, 9, 10, 11, 12].

ProSense PTD25-20-VH: economical industrial monitoring

This transmitter is a practical choice for confirming suction and tracking substantial changes. It has a 1/4-inch male NPT process connection and an M12 electrical connector; purchase the mating cable separately.

The vacuum output direction is important: 4 mA corresponds to 0 PSIG, and 20 mA corresponds to −14.5 PSIG. The installation manual lists 145 PSI permissible positive overload for this model. That is a survival specification, not a calibrated positive-pressure measuring range [9].

The product page gives a conservative accuracy specification of ±0.75%. Applied to the full vacuum range, this is about ±0.75 kPa. The detailed installation manual lists a tighter model-specific characteristic

deviation; use the conservative figure unless the supplier confirms the specification for the delivered instrument [8, 9].

Selection judgment: suitable for basic vacuum monitoring; it may be too coarse for small differences in weak suction.

ProSense EPS25-V14-1001: commissioning and diagnostics

This compound-pressure sensor provides a local display, analog output, and switching functions. It can distinguish normal vacuum from an abnormal positive-pressure condition. The process connection is 1/4-inch male NPT; an M12 cable is required.

The catalog lists 145 PSIG proof pressure. Its characteristic deviation is ±0.5% of span under the stated limit-setting convention, approximately ±1 kPa over the full roughly 200 kPa compound span [11].

Selection judgment: useful for commissioning, direct readings, and thresholds. A narrower-range instrument is preferable when the task is resolving a small change in pickup vacuum. Rescaling the analog output does not automatically improve the underlying sensor accuracy.

Omega PX309-015V5V: better stated static accuracy

The Omega option has a 1/4-inch male NPT connection, attached cable, and ±0.25% BSL static accuracy for combined linearity, hysteresis, and repeatability. On its nominal 15 PSI vacuum range, this corresponds to

$$0.0025(15) = 0.0375 \text{ PSI} \approx 0.26 \text{ kPa}. \tag{4}$$

This static-accuracy figure is not the complete temperature-inclusive error budget [12, 13].

Selection judgment: a useful choice among these candidates for quantitative vacuum comparisons with a voltage output. Order the exact `PX309-015V5V` variant; similarly named compound versions can have different output polarity. Its specified overload does not support direct connection to a 100 PSI supply.

5.2 Candidates for S_1: discharge pressure and bag restriction

Table 5. Low-pressure candidates for the discharge/collector channel.

Candidate	Measuring range	Output / supply	Price
ProSense `PTD25-20-0100WCH`	0–100 inH_2O; approximately 0–24.9 kPa or 0–3.61 PSIG	4–20 mA. 9.6–32 VDC supply.	$176
ProSense `DPTA-20-25`	0–25 inH_2O differential; approximately 0–6.23 kPa	4–20 mA. 12–36 VDC supply.	$160

The PTD model measures positive gauge pressure only. It offers a threaded industrial connection and a

wider low-pressure range. Its listing specifies ±0.75% accuracy [14]. **It is the preferred practical starting candidate for S_1 in this assessment.**

The DPTA is useful after confirming that the pressure difference stays within 6.23 kPa. Its ±1% full-range specification corresponds to about ±62 Pa. It is intended for clean, dry, noncorrosive gas and has a 15 PSIG maximum static line/proof rating. Keep it away from any possible 100 PSI exposure [16]. The product listing was backordered when checked [15].

For bag pressure relative to atmosphere, connect the DPTA high port to the upstream bag pressure tap and the low port to atmosphere. For an enclosed filter's pressure drop, connect to opposite sides of the filter. This 25 inH_2O model should not be chosen for S_2 without first confirming that maximum suction remains within its range.

5.3 Electrical integration and procurement details

These replacements are not all direct substitutes for the photographed 5 V-powered transmitter. The existing S_P likewise retains its own, presently unspecified interface.

- Provide each selected sensor's required supply voltage.
- Use an appropriate current-input interface for 4–20 mA models.
- Scale voltage outputs to the permitted analog-input range.
- Verify the exact connector pinout, process thread, and mating cable.
- Update pressure conversion, zero, full-scale value, and output direction in software.

A controller with a 3.3 V analog input requires suitable conditioning for 0–5 V or 0–10 V signals. Pressure-range compatibility alone does not establish electrical compatibility.

6 A controlled single-versus-tandem comparison

The following is a proposed bench procedure, not a report of completed testing. Use the existing S_P for supply-pressure logging throughout.

6.1 Prepare two meaningful configurations

Configuration A: one powered Line Vac, with the other physically removed and replaced by a smooth connector of suitable bore.

Configuration B: both Line Vacs operating in the tandem arrangement.

Turning one unit off while leaving it installed retains its passive restriction. That can be a useful operating check, but it is not the same baseline as a properly assembled single-unit system.

6.2 Keep the important conditions consistent

Use the same nozzle geometry, nozzle-to-debris spacing, hose dimensions and routing, elevation, debris material and size distribution, starting mass, and bag/filter condition.

Record S_P under flow in every trial. If its tap is upstream of important restrictions, account for that limitation when describing the motive pressure delivered to the active unit or units. If the available supply cannot maintain the intended operating condition with two units, record that outcome as part of the practical comparison.

Table 6. Measurements for the comparison, using the existing supply sensor.

Measurement	**What it establishes**
Existing S_P under continuous flow	Supply pressure at its actual location and changes when the second unit operates.
S_2 during pickup	Suction-side operating pressure.
S_1 during pickup	Discharge pressure associated with downstream resistance.
Time to collect a known mass	Actual debris collection performance.
Pickup airflow in a clean-air trial, if measurable	Whether additional air enters through the nozzle.
Compressed-air consumption, if measurable	Air demand associated with the collection improvement.

Use several repeat trials and compare both clean and representative loaded-filter conditions. A useful collection metric is

$$\dot{m}_d = \frac{m_{\text{collected}}}{t}, \tag{5}$$

reported with the debris description, nozzle setting, pressure readings, and collector condition.

Measure pickup airflow rather than relying on exhaust flow: the second unit injects supply air, so more exhaust can occur without a corresponding increase in intake airflow.

A brief closed-pickup check, with discharge open and all components within their ratings, can establish a maximum-suction endpoint. At that endpoint, pickup airflow is zero; it does not demonstrate collection performance. Do not deliberately block the discharge to conduct this check.

6.3 Interpret pressures together with collection results

Table 7. Diagnostic interpretations to investigate; none is a diagnosis from photographs alone.

Observed result	Interpretation to investigate
Two units collect substantially more debris with adequate delivered motive pressure	A useful tandem benefit may exist.
More exhaust with little collection improvement	Much of the added flow may be injected supply air.
S_P falls when both units operate	The available supply or upstream delivery path may limit the tandem arrangement.
S_P remains steady, but it is upstream of restrictive components	Downstream pressure loss remains possible; inspect the tap location and delivery path.
S_1 rises as the bag loads and pickup deteriorates	Collector or downstream resistance is becoming important.
S_2 becomes more negative while collection decreases	A pickup or hose restriction may be reducing useful airflow.

7 Recommended revised configuration and remaining inputs

For the first revision, use the following arrangement:

1. **Retain the existing supply sensor S_P.** Verify its specifications and tapping location, and log it with the Line Vac operating.
2. **Establish a one-Line-Vac baseline near the pickup.** Use a smooth, adequately sized discharge route and compare it with the tandem assembly.
3. **Use a vacuum-capable instrument at S_2.** Select the ProSense `PTD25-20-VH` for economical monitoring, the `EPS25-V14-1001` for a local display and diagnostics, or the Omega `PX309-015V5V` when better stated static accuracy is the priority.
4. **Use a low-pressure instrument at S_1.** The ProSense `PTD25-20-0100WCH` is a practical initial candidate for discharge monitoring.
5. **Use an adequately vented collection bag or filtered receiver.** Size its flow passage and filtration area for all conveying air, and evaluate the effect of filter loading.

The confirmation of S_P changes the instrumentation recommendation: a supply-pressure sensor is already present, so neither photographed transmitter needs to be reassigned to that role. The assessment of the vacuum and discharge channels remains separate.

The inputs still needed for final sizing are the exact Line Vac model and bore, sustained available compressed-air flow, hose lengths and diameters, nozzle dimensions, lift and bends, collector/filter resistance, debris size and density, and required collection rate. The exact model and location of S_P would also make its operating data easier to interpret.

The recommended next engineering step is to correct the unresolved measurement ranges and flow

restrictions, then use the existing supply measurement and controlled collection trials to decide whether the closely coupled second Line Vac provides a worthwhile benefit.

8 Research literature and the master's thesis opportunity

8.1 Research motivation and defensible knowledge gap

The apparatus provides a tractable mechanical-engineering research platform because ejector performance, nozzle entrainment, pneumatic transport, and collector resistance are coupled. The main engineering question is not simply which configuration gives the most negative value at S_2. It is which configuration *collects a defined mass of the intended debris, at a specified recovery requirement, with acceptable compressed-air consumption and repeatability.* The compressed-air sensor S_P is already installed (though absent from the supplied figure); it should be logged throughout the proposed studies. The pressure range and location of S_P must be verified before using its reading as the delivered ejector motive pressure.

The papers summarized below establish that commercial ejector performance, gas–solid entrainment, multiple air-amplifier arrangements, transport losses, filtration and control have each been investigated. Consequently, neither "using an ejector to vacuum powder" nor "putting two ejectors in series" is a sufficient novelty claim. A thesis contribution must instead specify a **new, reproducible operating envelope, predictive model, design rule, diagnostic method, or experimentally validated control improvement** for the selected debris and geometry. The exact EXAIR Line Vac model and the intended debris distribution remain to be confirmed; results from other devices cannot be directly transferred without validation.

8.2 Verified journal publications mapped to this apparatus

Table 8. Peer-reviewed research foundations, proposed relevance, and transferability limitations.

Publication	Finding relevant to the study	Limit when applied to H2C
Xu, Liu and Pang (2016) [18]	Compare single-/double-Venturi powder ejectors experimentally and with CFD–DEM; examine nozzle position and deposition.	Double-Venturi *geometry within an ejector* is not equivalent to two commercial Line Vacs placed close together.
Lee *et al.* (2023) [19]	Analyze serial mass-flow amplifiers; show that the geometry and flow split influence amplification.	The numerical study examines a gas-flow amplifier arrangement, not H2C's particle pickup and bag loading.
AbdEl-hamid *et al.* (2018) [20]	Experimentally compare air–solid ejector mixing/tail-section geometries and operating conditions.	Results refer to different internal ejector geometries; use for mechanisms and test variables, not predicted H2C performance.
Ebrahimi, Crapper and Ooi (2016) [21]	Validate a CFD–DEM approach against gas and particle velocities in a conveying line, identifying quantitative discrepancies.	Their 6.5-m test line and particles differ from the short H2C suction system.

Publication	Finding relevant to the study	Limit when applied to H2C
Saleem *et al.* (2012) [22]	Show how filtration velocity and loading affect pressure drop and deposited dust-cake properties.	Pilot-scale pulse-jet bag filters are not the same as a small passive collection bag.
Mittal and Mallick (2025) [23]	Use pressure-signal characteristics to investigate incipient blockage in dense-phase conveying.	Dense-phase signals and thresholds cannot be adopted directly for an ejector-driven short line.
de Freitas *et al.* (2022) [24]	Develop an operating-condition selection method for air-pressure/airflow and energy use in a pneumatic conveyor.	Demonstration uses a blow tank and a much longer line; a different apparatus requires a new calibrated map.
Ishiharada *et al.* (2021) [25]	Examine how suction-nozzle and injection-pipe insertion lengths influence powder transport/loading ratio.	Their nozzle includes an injection pipe; use their dimensional-analysis approach, not their optimum dimensions.

Suggested literature-review structure for a student: begin with ejector fundamentals and commercial performance curves; then review gas–solid pickup, suction nozzle design, particle and bend losses, filter loading, and system efficiency. For each paper, document the apparatus dimensions, material, operating pressure, measured response, simulation/measurement uncertainty, and limitations relevant to H2C. Do not treat CFD-predicted performance or a published optimization result as validation of this particular prototype.

9 Five candidate mechanical-engineering master's thesis topics

Each of the following is a *separate* potential thesis, rather than a set of five required research tasks. Topic A is the most direct use of the existing equipment. Topic C is a focused experimental alternative. Topic E is most appropriate after an operating map and repeatable pickup have been established. These proposed hypotheses require testing; they are not findings.

9.1 Topic A — Single versus tandem Line Vacs: recovery and specific air use

Proposed title: *Experimental Characterization and Compressed-Air-Use Optimization of Single and Tandem Ejectors for Small-Debris Collection.*

Research question. Under what motive-pressure, nozzle and downstream-resistance conditions does operating both nearby Line Vacs increase actual debris recovery enough to justify the additional compressed-air requirement?

Testable hypothesis. A tandem arrangement may increase debris capture or transport under some conditions, but the benefit will depend on collector resistance and available motive flow. When the two units are close together and the collector resistance is substantial, a properly configured single unit may have a lower specific air requirement at equal acceptable recovery.

Independent variables. Ejector configuration (one or two); feasible delivered motive-pressure level; and controlled collector-resistance state. A secondary study could vary inter-ejector separation or discharge-bend radius, but this should not be added until the primary experiment is complete. The same material and nozzle configuration must be held constant across paired trials.

Method. Establish a true single-unit configuration by replacing the removed unit with a smooth connector of appropriate bore; do not simply disable one ejector and leave its internal passage installed. Maintain matched hose length, outlet connection, nozzle height and debris presentation as closely as possible, documenting any unavoidable change. Compare:

1. **Equal delivered motive pressure** at each active unit, where the air system can maintain that condition. This tests the practical benefit of adding a unit at a common pressure, but the *total* motive air use need not match.

2. **Equal total motive-air flow or air budget**, over the feasible overlapping range. This tests whether the tandem arrangement provides superior collection with the same air expenditure, and may require independent regulators/valves and calibrated motive-air flow measurement.

Report cases that cannot maintain the target pressure as supply-limited results rather than silently treating them as pressure-matched trials.

Measurements. Use S_P under flow; measure S_2 and S_1 with appropriately ranged instruments; weigh the debris before and after each trial; measure time, motive air consumed, and clean-air pickup flow where practical. For flow-based comparisons, verify the delivered pressure at the inlet of each active ejector or characterize restrictions between S_P and those inlets.

Analysis and contribution. Construct collection-rate, recovery-fraction, and specific air-consumption response surfaces with uncertainty intervals. Identify a practical operating envelope in which the second ejector is beneficial, neutral or detrimental. A negative result demonstrating the superiority of one ejector within defined conditions is a valid and publishable design finding. Published double-Venturi and

serial-amplifier results motivate the comparison but do not predict it [18, 19, 20].

Feasible master's scope. One debris surrogate, one nozzle, two configurations, three pressure levels and three collector states for the main matrix; reserve spacing/geometry changes for a bounded follow-on study.

9.2 Topic B — Nozzle geometry, pickup gap, and validated entrainment modeling

Proposed title: *Design and Experimental Validation of Pickup-Nozzle Geometry for an Ejector-Driven Debris Collector.*

Research question. How do nozzle entrance geometry, effective throat area and distance from the surface affect recovery, clean-air entrainment, and deposition for a fixed air budget?

Testable hypothesis. An inlet transition optimized for the chosen particle-size distribution and a controlled stand-off distance can improve recovery without increasing motive-air consumption. An overly restrictive nozzle may make S_2 more negative while *decreasing* useful pickup airflow.

Proposed hardware. Fabricate a small set of interchangeable nozzle inserts: a plain circular opening, a radiused/bell-mouth opening, and a rectangular slot. Compare geometries at matched open area where feasible; retain the same downstream system and one Line Vac. A simple adjustable fixture should maintain stand-off height, yaw and pitch relative to the debris surface.

Measurements and model. Obtain recovery, collection rate, S_2, motive flow and inlet air velocity or clean-air induced volume flow for each trial. Develop a clean-air CFD model of the nozzle/suction section with a *measured* downstream pressure/flow boundary condition. Validate the model against measured pressure and flow over an operating range; only then assess whether a limited Eulerian–Lagrangian or CFD–DEM particle model is needed. Mesh independence and sensitivity to turbulence closure should be reported, not assumed.

Contribution. A validated geometry recommendation and a dimensionless design map using stand-off-to-nozzle size, entrance area, and particle-size-to-opening ratios. Relevant nozzle and gas–solid transport precedents are available [25, 20, 21]. Do not claim that these precedents establish the optimum for H2C.

Feasible master's scope. Three nozzles, two stand-off distances and three feasible pressure settings, with replicated debris tests and a clean-air CFD model. Full multiphase CFD–DEM is optional, not a prerequisite.

9.3 Topic C — Filter loading, discharge resistance, and service criterion

Proposed title: *Experimental Modeling of Collector-Loading Effects on Pickup Performance in a Pneumatic Debris Collection System.*

Research question. How does accumulated material alter a collector's air resistance and reduce recovery, and can a measured pressure-difference trend define a reliable servicing or replacement condition?

Testable hypothesis. At controlled motive conditions and nozzle geometry, increasing collector resistance reduces induced suction airflow and eventually reduces debris recovery. Effective media area and loading history modify the shape of that degradation curve.

Independent variables. Effective filter-media area (e.g., two verified areas using the same medium), mass loading or standardized resistance state, and a modest set of operating pressures. Keep the material

and particle-size distribution fixed during model development.

Measurement boundary. Distinguish S_1 (discharge-side pressure at its tap) from *filter pressure drop*, which requires pressure measurements immediately upstream and downstream of the media. If an added differential sensor is used, verify its static pressure and dust-protection ratings. Determine filter face velocity from the total *actual* volumetric airflow at the filter inlet divided by active filtration area; this airflow includes entrained pickup air *and* the ejectors' motive air. Different flow quantities must not be mixed without converting them to a common state.

Experimental method. Begin with a clean, unladen medium. Add standardized portions of debris, record the filter pressure difference, S_P, S_1, S_2, collection performance and air consumption, then repeat across independent loading cycles. Treat checkpoints inside one loading cycle as correlated repeated measures, not independent replicates. Model the dependence of resistance on loading and face velocity and test a withheld loading cycle.

Contribution. A validated collector-pressure-loss model and a service or filter-area criterion tied to a specified recovery threshold, with uncertainty. Bag-filter cake research demonstrates that filtration velocity can change cake resistance, but a small unpulsed bag may behave differently [22].

Feasible master's scope. Two media areas, a single debris material, one nozzle and one ejector, with multiple full loading cycles. This offers a strong thesis without needing to optimize internal ejector geometry.

9.4 Topic D — Fault diagnosis from pressure, airflow, and recovery

Proposed title: *Experimental Development of Pressure and Airflow Diagnostics for Faults in an Ejector-Driven Debris Collector.*

Research question. Can a small, physically interpretable sensor set distinguish normal collection, partial inlet obstruction, an intake-hose leak and increased collector resistance?

Testable hypothesis. Joint features from S_P, S_2, S_1 and an independent flow/recovery reference can distinguish at least some faults more reliably than an absolute threshold applied only to the suction pressure.

Fault protocol. Impose a calibrated set of restrictions, controlled leak openings and repeatable filter-resistance states. Record ground-truth conditions from the imposed fault fixture plus weighing and flow measurement. Include healthy runs at different motive-pressure settings, because an operating-point change should not be misclassified as a fault.

Data and inference. Start with mean gauge pressures, slopes, pressure ratios and run-to-run variance. Add band-limited transient features only after checking the sensor response time and sampling rate. Compare interpretable thresholds or a state machine with one bounded classifier, and evaluate a confusion matrix, false alarms and detection latency on entire runs or days withheld from training. In particular, do not randomly mix time windows from the same run between training and test sets.

Contribution. A sensor-necessity assessment and reproducible diagnostic decision rules. Dense-phase blockage work motivates pressure-signal features, but the flow regime is different and published feature thresholds are not directly transferable [23].

Feasible master's scope. Four operating states (healthy plus three faults), two pressure levels, two debris-loading conditions, and repeated labeled runs. A complex neural-network model is unnecessary

unless it consistently outperforms transparent baselines.

9.5 Topic E — Feedback regulation and reduced compressed-air consumption

Proposed title: *Feedback Regulation of Motive Pressure for Reduced Compressed-Air Consumption in Pneumatic Debris Collection.*

Research question. Can automatic adjustment of motive pressure preserve a prescribed collection performance while reducing the standard volume of compressed air used per mass recovered as the collector becomes more restrictive?

Testable hypothesis. A sensor-driven pressure schedule with a bounded PI correction may reduce air use relative to a conservative high-pressure setting, and in some conditions relative to a fixed pressure selected for a clean collector. Whether it outperforms the *best feasible constant-pressure setting* must be established experimentally.

Actuation and model. Use a rated controllable regulator with known command/fail behavior. Perform steady tests at several pressure setpoints and resistance states to estimate a static map from measured delivered pressure to induced airflow, recovery and motive-air consumption. With one regulator feeding a fixed Line Vac, motive pressure and flow are coupled; do not claim independent control of both unless separate actuators exist.

Control design and comparison. Start with an open-loop pressure schedule; implement bounded PI feedback with anti-windup, filtered measurements, and safe pressure limits only after verifying actuator authority and response speed. Compare (i) a feasible conservative high fixed pressure, (ii) the best constant pressure found during characterization, and (iii) the scheduled/feedback approach on comparable debris and collector-loading sequences.

Outcomes and contribution. Report recovery constraints, air consumed per kilogram, collection time, steady-state pressure error and robustness to loading. Electrical *energy* reduction must not be claimed from air-volume savings alone; it requires measured compressor electric power or a validated, explicitly bounded compressor model. Research on pressure/airflow selection for other pneumatic conveyors provides a method precedent rather than direct H2C controller gains [24, 28].

Feasible master's scope. One ejector, one nozzle, three resistance states, a mapped regulator and one practical control method. More advanced MPC is optional after the basic schedule and PI comparisons are established.

9.6 Choosing a thesis with an achievable scope

Table 9. Recommended thesis selection based on current apparatus readiness; qualitative judgments are planning assessments, not experimental results.

Topic	Main research strength	Principal added requirement	Scope recommendation
A	Directly decides one versus two units	Flow/air-use measurement and a fair rebuildable baseline	**Preferred first thesis**; strongest immediate application
B	Geometry and flow-physics design	Repeatable fixture, interchangeable nozzles; CFD validation	Good fluid-mechanics option
C	Collector performance and service-life model	Filter differential-pressure measurement and complete loading cycles	**Focused experimental alternative**
D	Condition monitoring and diagnosis	Controlled fault hardware and correctly split validation data	Good mechatronics/instrumentation topic
E	Feedback and air-use efficiency	Rated pressure actuator, validated performance map	Best after baseline experiments

10 Common experimental methodology for the thesis options

10.1 Definitions, measurands, and performance criteria

A negative suction-pressure reading alone cannot define collection success. Use a primary gravimetric recovery measurement with carefully specified experimental duration and material presentation:

$$\eta_{\text{recovery}} = \frac{m_{\text{collected}}}{m_{\text{offered}}}, \qquad \dot{m}_{\text{collection}} = \frac{m_{\text{collected}}}{t_{\text{run}}}, \tag{6}$$

$$V_{\text{motive,std}} = \int_0^{t_{\text{run}}} Q_{\text{motive,std}}(t)\,dt, \qquad C_{\text{air}} = \frac{V_{\text{motive,std}}}{m_{\text{collected}}}. \tag{7}$$

Here, m_{offered} is the debris mass made available to the pickup, $m_{\text{collected}}$ is mass retained in the collector within the run, t_{run} is run time, and $Q_{\text{motive,std}}$ is *measured* motive-air flow converted to a declared standard condition. C_{air} is therefore standard air volume per kilogram collected, not electrical compressor energy. State how debris remaining on the surface, lodged in the nozzle/hose, and retained by the filter is handled in the mass balance. A fixed-time study and a fixed-target-mass study answer different questions; use one as the primary protocol and report the other separately.

When analyzing filters, define

$$\Delta p_f = p_{f,\text{upstream}} - p_{f,\text{downstream}}, \qquad u_f = \frac{Q_{f,\text{actual}}}{A_{\text{media,active}}}, \tag{8}$$

where v_f is the mean filtration face velocity. Its airflow and area must correspond to the same physical control surface. Do not use the compressed-air supply SCFM alone as the filter's local volumetric flow.

10.2 Instrument plan, reference measurements, and calibration

Table 10. Minimum data channels to support a quantitative master's thesis.

Quantity	Instrument or method	Verification requirement
Supply pressure	S_P, already present	Confirm range, gauge/absolute reference, tap position and live pressure with flow.
Pickup pressure	Vacuum-rated S_2 and protected static tap	Atmospheric-zero check, range and response-time verification.
Discharge pressure	Low-range S_1; add separate filter differential if needed	Confirm positive pressure within rating; distinguish hose losses from filter-only loss.
Motive air use	Calibrated flow sensor/flow meter or validated industrial air-meter method	State standard reference conditions and uncertainty.
Pickup-induced airflow	Clean-air baseline flow measurement or calibrated inlet test device	Separate induced flow from injected motive flow; check device restriction.
Recovery and deposition	Weigh preconditioned offered/collected/retained debris	Calibrated balance, repeatable dosing, moisture and leakage controls.
Collector state	Measured filter loading and pressure differential	Defined clean/loaded conditions; repeatability across independent cycles.

Provide common time stamps and log regulator setting, commanded state, valve condition, temperatures if material, run ID, filter state, nozzle gap, and any audible or visible blockage. At each test day, check sensor zero and pressure/flow readings against a suitable reference; use a calibration curve rather than assuming linear response beyond a sensor's specified range. Express combined uncertainty following the JCGM *Guide to the Expression of Uncertainty in Measurement* [27]. Establish acceptable uncertainty for the *difference* between configurations, not merely for individual readings.

10.3 Proposed initial experiment matrix for Topic A

Use a screening pilot to establish safe and feasible delivered pressure values and representative resistance states. One defensible initial balanced matrix is:

$$N = 2\,\text{configurations} \times 3\,\text{pressure levels} \times 3\,\text{collector states} \times 5\,\text{independent repeats} = \boxed{90\ \text{debris trials}}. \tag{9}$$

The count is a starting workload estimate, **not a statistical power calculation**. Adjust the number of repeats after evaluating pilot variance and a practically important minimum difference. Some conditions may be infeasible with the available compressed-air supply; report these feasibility bounds rather than forcing nominal settings. If stable air-delivery pressure differs across configurations, separate actual-delivered-pressure comparisons from equal-upstream-gauge comparisons.

Randomize configuration and pressure-order within practical *blocks* (for example, test day, fixed material batch or filter conditioning). Standardize preparation between trials. Compare the average effects and interaction terms with confidence intervals using a factorial/blocking model or equivalent mixed-effects analysis, and inspect residuals. Treat repeated readings within the same debris run and checkpoints within the same filter-loading cycle as *correlated* observations. NIST's experiment-design guidance covers

randomized blocks, factorials and response-surface extensions [26].

10.4 Validation, uncertainty, and completion criteria

The student should declare, *before* the final experiment, a minimum acceptable recovery fraction, relevant debris class, maximum collection time and acceptable motive-air use. These are project requirements to be selected with the advisor; the apparatus photographs do not establish suitable numerical thresholds.

Report (i) the mean and uncertainty of each response, (ii) effect estimates and intervals between configurations, (iii) retained/deposited mass and unaccounted debris, and (iv) a validation campaign on conditions not used to fit any model. For Topic B or E, reserve distinct test combinations for validation; for Topic C reserve whole filter-loading cycles; for Topic D reserve full runs or days. Consider the complete measurement system, including flow-meter pressure-drop effects, supply fluctuation, balance repeatability and filter state, in the uncertainty budget [27, 26].

For any claim of electrical-energy improvement, add compressor input-power measurements and calculate

$$E_{\text{elec}} = \int_0^{t_{\text{run}}} P_{\text{compressor}}(t)\, dt, \qquad e_{\text{spec}} = \frac{E_{\text{elec}}}{m_{\text{collected}}}. \tag{10}$$

Declare whether the compressor, dryer, controls, reservoir losses and standby periods belong to the system boundary. U.S. DOE compressed-air guidance supports separating supply-system efficiency from pressure and flow at the pneumatic user [28].

10.5 Safety and apparatus protection

Use only pressure-rated regulators, hoses, fittings, tubing, isolation valves and properly restrained connections suitable for the maximum compressor pressure. Never pressure-test a low-range vacuum or differential sensor by exposing it to the 100-PSI motive line. Ensure the collector is vented/filter-rated for the combined ejector-plus-pickup airflow, and prevent excessive pressure accumulation in the bag/receiver. Add accessible isolation and emergency shutoff measures, inspect leaks, and consider noise protection. Characterize the debris for toxicity, respirability, combustibility and electrostatic hazards *before* testing; do not begin with hazardous dust or unknown fine powder. These controls must be set through the laboratory's formal risk assessment, not inferred from the design photograph.

11 Proposed 12-month master's research schedule

Table 11. A realistic planning sequence for one selected topic; timing is illustrative and depends on laboratory access.

Months	Primary work	Gate / tangible deliverable
1–2	Define research question, survey literature, characterize the debris, verify Line Vac model and S_P tap	Approved scoped proposal, measurement requirements, preliminary hazard analysis
3–4	Complete apparatus modifications, calibrations, sampling plan and pilot trials	Working data-acquisition system; repeatability, feasible pressure range and estimated variance
5–7	Conduct randomized main experimental campaign and document any failures	Versioned test protocol and reproducible raw/processed data set
8–9	Fit interpretable engineering model, quantify uncertainty and compare alternatives	Response surfaces, model assumptions and uncertainty budget
10	Run independent validation tests under withheld operating points or test days	Performance bounds and verified contribution statement
11–12	Write and revise thesis; prepare journal/conference manuscript and defense	Completed thesis, documented hardware/software and presentation

Suggested initial thesis selection

Choose Topic A if the immediate project decision is whether the existing two-Line-Vac arrangement offers a meaningful improvement over a single unit. It uses the present test rig most directly and creates a measured system operating map. **Choose Topic C** if the research should concentrate on a narrower, statistically well-controlled filtration problem. Topics B, D and E can follow once pressure sensing and clean-air/debris baselines are repeatable.

A defensible thesis need not demonstrate that two units are better. A measured finding that a single ejector achieves similar or greater recovery with less compressed-air use can yield a valuable design rule, provided the experimental comparison is fair and the uncertainty is quantified.

References: Journal Research, Methods, Hardware and Apparatus

Reference classification. The 28 sources comprise eight verified journal publications, three standards/method resources, 16 manufacturer/supplier resources, and the four-page user-provided H2C apparatus source (internal project document, not peer reviewed). The journal references establish scientific precedents; supplier literature supports specifications and purchasing. The `.tex` document uses a built-in bibliography and diagrams, with no external `.bib` or graphics files. Its `needspace` handling is optional for compatibility with TeXstudio installations that lack that package.

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